Kai Liu

Contact information

Dr. Yvette Kanouff School of Data,
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Current Affiliation

Postdoctoral scholar in Dr. Yvette Kanouff School of Data, Mathematical, and Statistical Sciences (SDMSS), University of Central Florida, August 2025 to present. Research area in Applied Harmonic and Functional Analysis, Quantum Information and Quantum Computing.

Researchers IDs

Google Scholar
ORCID
Scopus Author ID
GitHub Profile

Education

- Ph.D. in Mathematics, University of Central Florida (UCF), Fall 2019 to Summer 2025, GPA 3.929/4.0.
  Dissertation: Phase retrieval and fidelity preservation of quantum channels.
  Ph.D. Advisor: Dr. Deguang Han.
  Dissertation Committee: Dr. Deguang Han, Dr. Dorin Dutkay, Dr. Jiongmin Yong, Dr. Qiyu Sun, Dr. Shengli Zou.

  Selected for UCF Trustees Doctoral Fellowship, UCF, 2019~2023.
  Winner of COS Dean's Doctoral Fellowship, College of Science, UCF, 2023.
  Winner of Research Excellence Award, Department of Mathematics, UCF, 2023.
  Winner of COS Dean's Stellar Doctoral Fellowship, UCF, 2025.
- M. S. in Mathematics, University of Central Florida (UCF), Fall 2019 to Spring 2021, GPA 3.917/4.0.
- M. S. in Mathematics and Applied Mathematics, Beijing Institute of Technology (BIT), Fall 2016 to Spring 2019, GPA 3.545/4.0.
  Dissertation: Operator Weighted Concurrence Measure on Two-Qubit Quantum System.
  M. S. Advisor: Dr. Lining Jiang.
- B. S. in Mathematics and Applied Mathematics, Beijing Institute of Technology (BIT), Fall 2012 to Spring 2016, GPA 3.42/4.0.
  Dissertation: A brief introduction to Clifford Algebra and its properties.

Academic Positions Held

- Postdoctoral Scholar, Dr. Yvette Kanouff School of Data, Mathematical, and Statistical Sciences (SDMSS), UCF. August 2025 Present
- Graduate Teaching Associate, Department of Mathematics, UCF. Spring 2022 to Summer 2025

- Graduate Teaching Assistant, Department of Mathematics, UCF. Fall 2019 to Fall 2021
- Graduate Teaching Assistant, Department of Mathematics and Statistics, BIT. Fall 2017 to Spring 2019

Research Interests

- Mathematical Foundations of Quantum Information --- Functional and harmonic analysis, operator theory and operator algebras, frame theory, and representation theory, quantum property preservation, and quantum tomography.
- Mathematical Physics and Quantum Control --- Open quantum systems, quantum dynamics and quantum control theory, quantum measurement theory, and mathematical modeling of quantum information processing in physical systems.
- Quantum Computing, Machine Learning, and Data Science --- Quantum computation and algorithms, quantum error correction and fault-tolerant quantum computation, quantum computational advantage, quantum machine learning, machine learning, and mathematical data science.

Research Activities

The following entries are listed in reverse chronological order (most recent first) by event date and by online posting date.

Publications, preprints and papers under review

Published Journal Articles

- Phase retrievability of frames and quantum channels, Kai Liu and Deguang Han, Linear Algebra and its Applications, Vol. 725, pp. 378--400, 2025. PDF
- Frame dimension functions and phase retrievability, Deguang Han and Kai Liu, Advances in Operator Theory, Vol. 9, Article 31, 2024. PDF
- Zero-error correctibility and phase retrievability for twirling channels, Deguang Han and Kai Liu, Reports on Mathematical Physics, Vol. 93, No. 1, pp. 87--102, 2024. PDF
- Orbit-injective covariant quantum channels, Kai Liu, Chuangxun Cheng, and Deguang Han, Linear Algebra and its Applications, Vol. 668, pp. 28--50, 2023. PDF
- Enhancing phase retrievability of quantum channels via interferometric coupling, Kai Liu, Deguang Han, and Omar Nour, Journal of Physics A: Mathematical and Theoretical, 59(44), 445301, 2026. DOI: 10.1088/1751-8121/aeb83a. arXiv PDF

Accepted Manuscripts

- Extreme Points of Optimal Dual Frames for Erasures, Shankhadeep Mondal, Deguang Han, and Kai Liu. Accepted for publication in Sampling Theory, Signal Processing, and Data Analysis.

Revised Manuscripts Under Review

- Fidelity preserving and decoherence for mixed unitary quantum channels, Kai Liu and Deguang Han, arXiv:2410.20117, 2024; revised 2026. Revision submitted; under review by Quantum Information Processing. PDF

Submitted Manuscripts and Preprints

- Vanishing Moment Dilations and Applications to Balanced and Graph Frames, Deguang Han, Shankhadeep Mondal, Kai Liu, and Omar Nour, arXiv:2610.11043, 2026. Under review by Journal of Fourier Analysis and Applications. PDF
- Frame phase retrievability and state distinguishability of quantum channels, Deguang Han and Kai Liu, arXiv:2608.30006, 2026. Under review as a book chapter. PDF

Ph.D. thesis

- Phase Retrieval and Fidelity Preserving Quantum Channels, Kai Liu, Ph.D. dissertation, University of Central Florida, 2025. PDF

Conference, workshop and colloquium presentations

- Invited talk. ``Frame phase retrievability and state distinguishability of quantum channels." Special Session on New Faces in Operator Theory, 2027 Joint Mathematics Meetings, Chicago, IL, January 12--15, 2027. Event
- Invited talk. ``Phase Retrievability of Quantum Channels and Interferometric Coupling." Special Session on Mathematical Aspects of Quantum Information and Computation, IV, 2026 Fall Western Sectional Meeting, Arizona State University, Tempe, AZ, November 8, 2026. Program
- Invited talk. ``Dynamical Sampling for Pure State Identification in Open Quantum Systems." Special Session on Recent Developments in Applied Harmonic Analysis, II, 2026 Fall Western Sectional Meeting, Arizona State University, Tempe, AZ, November 7, 2026. Program
- Invited talk. ``Operator valued frames and phase retrieval of quantum channels." 9th International Conference on Computational Harmonic Analysis (ICCHA 2026), in conjunction with the 38th Shanks Lecture, Vanderbilt University, Nashville, TN, May 21, 2026. Program
- Invited talk. ``Enhance Phase Retrievability via Quantum Interferometer." Special Session on Frames, Sampling, and Quantum Information Theory, II, 2026 Spring Southeastern Sectional Meeting, Georgia Southern University, Armstrong Campus, Savannah, GA, March 28, 2026. Program
- Contributed talk. ``Zero-error Correctibility and Phase Retrievability for Twirling Channels." AMS Contributed Paper Session on Computer Science, Information, and Communication, I, 2026 Joint Mathematics Meetings, Washington, DC, January 4, 2026. Program
- Invited talk. ``Frame Theory in Phase Retrievable POVMs and Quantum Channels." Special Session on Inverse Problems IV, 2025 Fall Central Sectional Meeting, Saint Louis University, St. Louis, MO, October 19, 2025. Program
- Invited talk. ``Phase Retrievability of Frames and Quantum Channels." AMS Special Session on Recent Progress in Frame Theory and Harmonic Analysis, I, 2025 Fall Western Sectional Meeting, University of Denver, Denver, CO, August 23, 2025. Program
- Invited talk. ``Fidelity Preservation for Mixed Unitary Quantum Channels." Workshop on New Classes of Spectral Duality, Inverse Problems, Operators, Analysis, and Algorithms, online, March 22--23, 2025.
- Invited talk. ``Frames and Generalized phase damping quantum channels." October 2024. Colloquium organized by the Department of Mathematics, UCF.
- Invited talk. ``Fidelity preserving phenomenon under mixed unitary quantum channels." April 19, 2024. Mathematics and Applications Seminar organized by the Department of Mathematics, UCF. Event

Current projects and working papers

- Quantum dynamical measurements and phase retrieval
- Post processing complexity of noncommutative quantum graphs

Professional Activities

Session organization

- (Organizer) Special Session on Frames, Sampling, and Quantum Information Theory, 2026 AMS Spring Southeastern Sectional Meeting, March 2026.

Journal reviewer

- Quantum Science and Technology; Journal of Physics A, Mathematical and Theoretical; Machine Learning, Science and Technology; Quantum Reports; Mathematics; Symmetry.

Reviewing Editor (Mathematics) in Springer Nature Reviewer Programme, 2026-Present

Honors, Fundings, Fellowships and Awards

- Preeminent Postdoctoral Program (P3) Professional Development allowance, Postdoc funding, UCF, Fall 2025. Funding amount: $5,000 USD.
- COS Dean's Stellar Doctoral Fellowship, an award for Ph.D students with outstanding progress towards degree completion, UCF, Spring 2025. Fellowship amount: $5,000 USD.
- COS Dean's Doctoral Fellowship, award for doctoral students who excel in both their academic accomplishments and their contributions to high-quality instrucion of UCF undergraduates, UCF, Fall 2023 to Spring 2024. Fellowship amount: $5,000 USD.
- Research Excellence Award, an award to honor graduate students of Department of Mathematics to conduct research of high quality, Department of Mathematics, UCF, Fall 2023.
- UCF Trustees Doctoral Fellowship, a multi-year award for outstanding graduate students, UCF, Fall 2019 to Spring 2023. Fellowship amount: $100,000 USD.

Teaching Experience

Courses Taught: College Algebra; Calculus 2; Calculus 3; Ordinary Differential Equations; Linear Algebra.

Teaching related services:

- Course Co-coordinator, MAP 2302, Ordinary Differential Equation — Fall 2025 to Summer 2027

  Responsibilities:

  - Syllabus and scheduling: Authorize the common syllabus, built the master schedule with lecture pacing, quiz/exam windows, and due dates.
  - Pearson online homework: Work directly with Pearson to create and maintain the MyLab master shell, sync rosters/gradebook, and resolve access or assignment issues.
  - Instructor communication: Coordinate across all section instructors throughout the semester, share updates.

Instructor of record:

- Postdoctoral scholar, UCF — Fall 2026 to Spring 2027

  MAP 2302, Ordinary Differential Equation. Section 0007 (Fall 2026)
  MAS 3105, Matrix and Linear Algebra. Section 0005 (Fall 2026)

- Postdoctoral scholar, UCF — Fall 2025 to Spring 2026

  MAP 2302, Ordinary Differential Equation. Section 0002, 0009 (Fall 2025)
  MAP 2302, Ordinary Differential Equation. Section 0002, 0010 (Spring 2026)

- Teaching associate, UCF — Fall 2024 to Summer 2025

  MAC 2313, Calculus 3. Section 0006 (Fall 2024)
  MAC 2312, Calculus 2. Section 0026 (Spring 2025)
  MAS 3105, Matrix and Linear Algebra. Section C003 (Summer 2025)

- Teaching associate, UCF — Fall 2023 to Summer 2024

  MAC 2313, Calculus 3. Section 0001, 0002 (Fall 2023); Section 0014 (Spring 2024)
  MAS 3105, Matrix and Linear Algebra. Section D003 (Summer 2024)
  MAP 2302, Ordinary Differential Equation. Section C004 (Summer 2024)

- Teaching associate, UCF — Fall 2022 to Summer 2023

  MAS 3105, Matrix and Linear Algebra. Section 0006 (Fall 2022)
  MAS 3105, Matrix and Linear Algebra. Section D001, D005 (Summer 2023)
  MAC 2313, Calculus 3. Section 0012 (Spring 2023)

- Teaching associate, UCF — Spring 2022 to Summer 2022

  MAC 2312, Calculus 2. Section 0018 (Spring 2022)
  MAC 2313, Calculus 3. Section C002, C007 (Summer 2022)

Teaching assistant experience:

- Teaching assistant, UCF — Spring 2021 to Fall 2021

  Course: MAP 2302, Ordinary Differential Equation; MAC 1105, College Algebra.

- Teaching assistant, BIT — Fall 2017 to Spring 2019

  Course: Mathematical Analysis , Linear Algebra.

Selected Coursework

Analysis, Real Analysis, Complex Analysis, Functional Analysis, Applied and Computational Harmonic Analysis, Advanced Linear Algebra and Matrix Theory, Abstract Algebra, Theory and Application of Semigroups, Topology, Riemann Geometry, Optimization, Theory, Mathematical Statistics, Bayesian Analysis, Scientific Computing, Applied Numerical Analysis, Machine Learning, Graph Theory, Financial Mathematics.

Computer proficiency

Python, Matlab, LaTeX, Qiskit.

References

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